

Adalida Baca

Doaa Omran Mohamed

English 250

March 26, 2019

Presence and Absence

Sun and moon, light and dark, presence and absence these are all perfect examples of a binary opposition. Binary oppositions are used in both disciplines of English and Computer Science. Computer scientists apply this concept to machine learning while literary critics use this to understand cultural concepts. Ferdinand de Saussure in his *Course in General Linguistics* coined this concept and Levi Strauss later applied it to mythology. In Strauss' *Structural Study of Myth*, he derives the concept of the mytheme showing that readers perform a breadth first search to understand these cultural concepts. Binary oppositions, coined by Saussure and developed by Strauss, uses a breadth first search on mythemes to understand cultural truths.

First, a binary opposition is two complementary terms that are opposed in meaning where one term helps understand the other. For instance, to understand dark we must understand it's opposite light because "a loan-word no longer counts as such whenever it is studied within a system; it exists only through its relation with, and opposition to, words associated with it" (Saussure 22). By placing these opposing terms against each other we can understand them better and synthesize more abstract concepts. Once one abstract concept is derived, we can repeat that

same process to a new and different concept causing the formation of what is considered a binary hierarchy in English, and a decision tree in machine learning.

These larger binary trees enable readers to understand and learn abstract concepts to its purest form. For instance, if a child is reading a story with a protagonist and antagonist, then they understand the protagonist is the hero which is good, and the antagonist is the villain which is bad. The child can understand the binary concepts of good and bad by first understanding the binary opposition between hero and villain. This is an example of an English binary hierarchy used in mythology and literature.

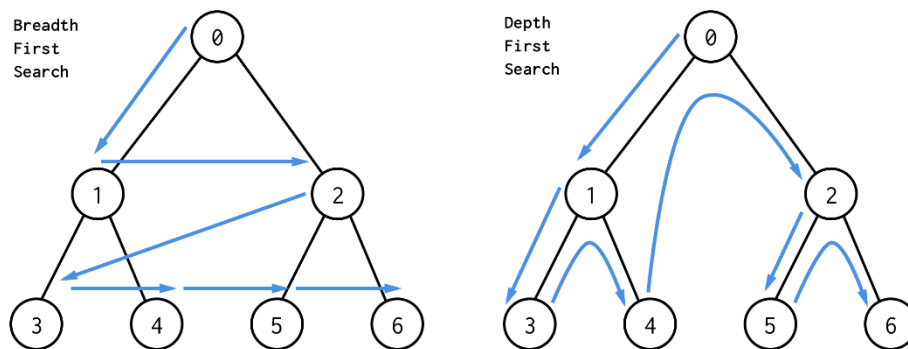
Machine learning uses this same concept except it's called a decision tree not a binary hierarchy. To illustrate, if a computer is running a program and is told to either run a program or not, it would use a decision tree to make that judgement. This is most effective because computers operate using binary values such as true and false or 0 and 1. If a decision tree is true, the computer runs or doesn't run the program and the same can be said for its inverse. It's clear that a binary opposition is a link that parallels English and Computer Science.

Ferdinand Saussure, a famous Linguist who influenced Structuralism, is credited with discovering this concept when studying language and semiotics. Saussure claimed that these opposing terms were not contradictory to one another but were complementary. For instance, he wrote that the "Relations and differences between linguistic terms fall into two distinct groups, each of which generates a certain class of values. The opposition between the two classes gives a better understanding of the nature of each class" (Saussure 123). It is necessary that two

opposing terms are viewed as complementary because their results return a value necessary to understanding human culture and values.

These values create and expand binary trees for human understanding. For instance, he uses these different binaries: parole and langue, synchronic and diachronic, and syntagmatic and paradigmatic to explain human understanding through language. These binaries communicate and “correspond to two forms of our mental activity, both indispensable to the life of language” (Saussure 123). This dissection of language illustrates that humans think in binary so much so that it influenced speech. If we know that speech is binary which leads to a hierarchy, we can learn to navigate and grasp these concepts better.

Levi Strauss, another famous Structuralist did just that in his *Structural Study of Myth*. He learned to navigate binary trees in language using a breadth first search algorithm. A breadth first search algorithm is a computer science concept used for navigating a decision tree and is opposite a depth first search.



A breadth first search algorithm is illustrated as moving from left to right and from top to bottom. This differs from a depth first search which traverses through the tree from top to bottom and left to right.

Stauss uses this technique when discussing how a myth is read and interpreted. He first explains a "*bundles of such relations*" (Strauss 431) is a collection of myth elements which illustrate hierarchies. Furthermore, he explains hierarchies are necessary for learning because "it is only as bundles that these relations can be put to use and combined so as to produce a meaning" (Strauss 431). In his later works, he refers to this concept as a Mytheme which is the collection of these bundled relations. Mythemes are used to perform a breadth first search much like a computer. The machine learns to group tasks together and perform these tasks based off what it learns from bundles of trees.

These bundles are the same concept but structured visually different because it categorizes mythemes. For instance, Strauss's myth is "treated as would be an orchestra score perversely presented as a unilinear series... the result is a chart" (Strauss 432). This chart is the same as a binary tree just in chart form used to find meaning. Strauss calls meaning "harmony" and finding harmony is the goal. To find this, "they would then find out that an orchestra score, in order to become meaningful, has to be read... from left to right" (Strauss 432). Strauss is using this chart to find harmony which is the same as using a binary tree to find meaning.

He reads this chart the same way a breadth first search is performed. For instance, in his chart, he claims that "to understand the myth, then we will have to disregard one half of the diachronic dimension (top to bottom) and read from left to right" (Strauss 433). The first half of the breadth first search (left to right) allows the reader to understand the meaning behind these binaries like good and bad.

To determine the beginning binaries, we must perform the second half (top to bottom) of the breadth first search. To illustrate, Strauss confirms that “were we to tell the myth, we would disregard the columns and read the rows from left to right and from top to bottom” (Strauss 433). Strauss is showing the second part of the binary which is telling the difference between binaries. For instance, when a child learns about good and bad, they have first learned about the hero and the villain. The second part of the algorithm is the lower concepts like hero and villain while the first part is the abstract concepts like good and bad.

The techniques and procedures performed to understand truth by English and Linguistic disciplines is parallel to that of Computer Science. Ironical, considering that one of those disciplines is STEM related and the other is Humanities related. In college we see these two fields pitted against each other as if they were opposites much like a binary. However, we have now proven with the use of binary oppositions and mythemes (hierarchy), that they are actually complementary. Both fields are trying to understand the same universal truths of presence and absence, light and dark, even the sun and the moon. Both fields are important in understanding the other.

Works Cited

Lévi-Strauss, Claude. "The Structural Study of Myth." *The Journal of American Folklore*, vol. 68, no. 270, 1955, pp. 428–444. *JSTOR*, www.jstor.org/stable/536768.

Saussure, Ferdinand De. "1959[1906-1911]. Course in General Linguistics, Translated by Wade Baskin, Selected 114–117, 120–122. New York: Philosophical Library." *The Discourse Studies Reader*, 2014, doi:10.1075/z.184.13sau.